Eight patents

What Mexico and Central America stopped buying, and what it cost them

— *a whole country's output of new ideas, 2024*

PART 1 OF 4 · THE SIGNAL

Its inventors filed eight patents all year.

Guatemala, 2024. Population 18.4 million.

8

PATENT APPLICATIONS, WHOLE COUNTRY, WHOLE YEAR

KEY

One dot is one patent application filed by someone living in the country.

not eight thousand.
eight.



1 DOT = 1 PATENT

Patent applications, residents, 2024 · World Bank WDI, IP.PAT.RESD · retrieved 17 Oct 2026

PART 1 OF 4 · THE SIGNAL

Korea filed this many.

Same year. Same chart. Same definition.

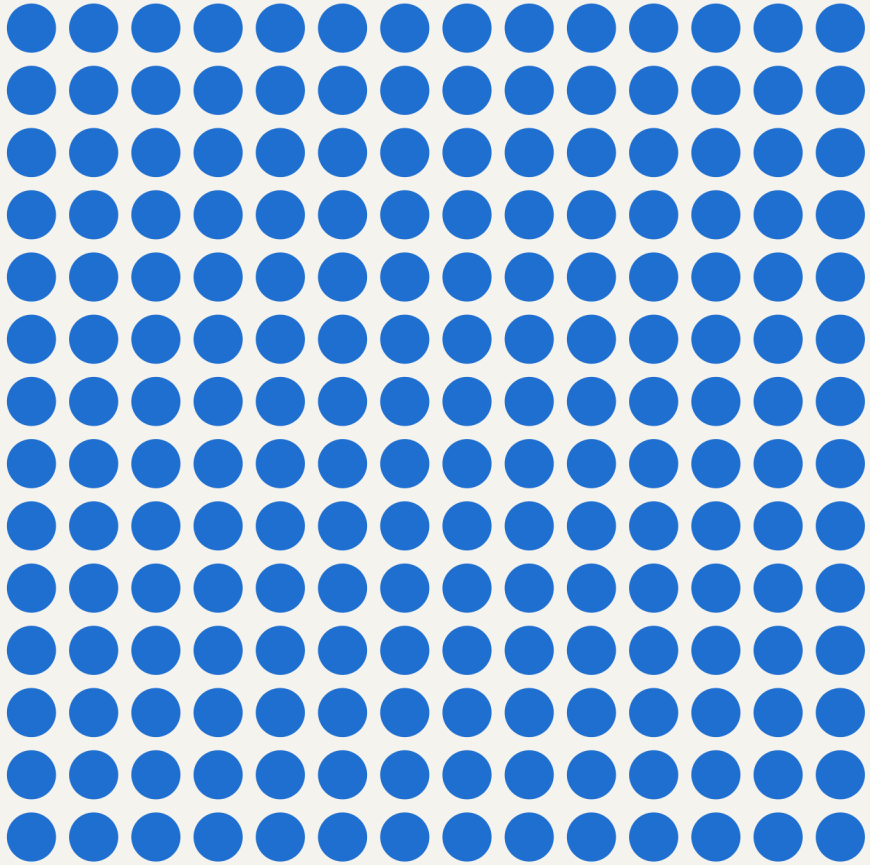
195,786

PATENT APPLICATIONS, SOUTH KOREA, 2024

KEY

Every dot is one thousand patents. On this scale Guatemala's eight would be one one-hundred-and-twenty-fifth of a single dot.

1 DOT = 1,000 PATENTS



and Korea has fewer people than Mexico does

Patent applications, residents, 2024 · World Bank WDI, IP.PAT.RESD · retrieved 17 Oct 2026

PART 1 OF 4 · THE SIGNAL

Per person, the gap will not fit on the slide.

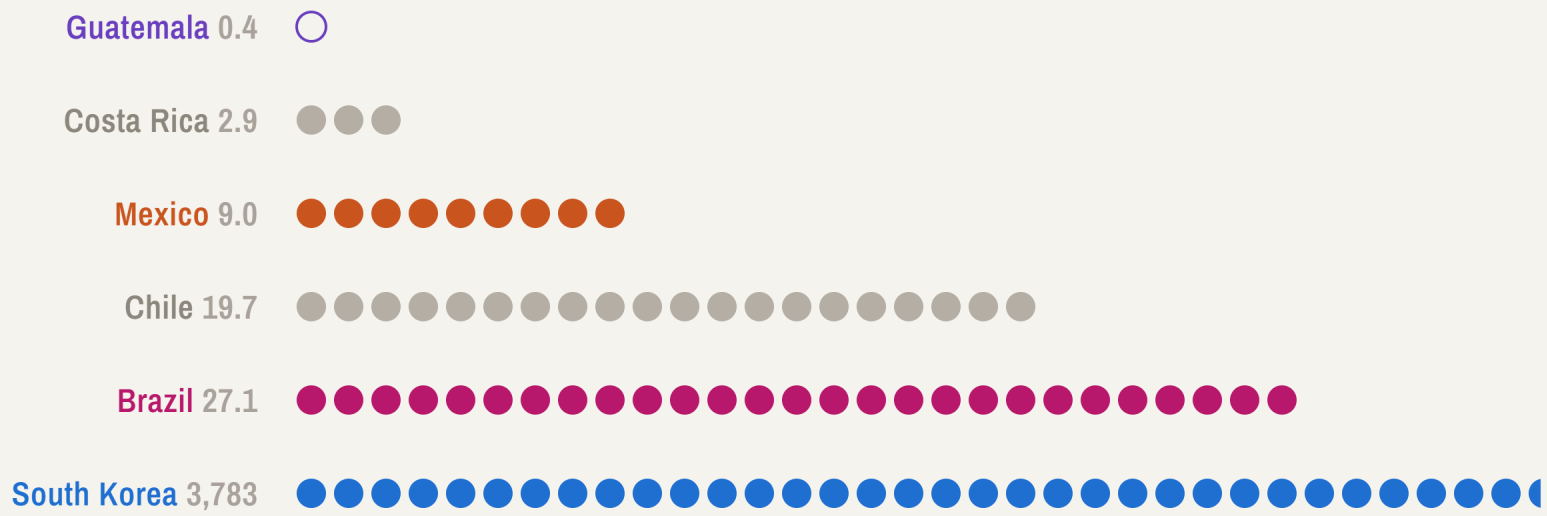
Patents per million people, 2024.

420×

KOREA'S FILINGS PER PERSON AGAINST MEXICO'S. AGAINST GUATEMALA, 8,704×

KEY

One dot is one patent per million people. Guatemala's is hollow because it is less than one.



Korea's row keeps going for another 55 slide widths

Patents by residents ÷ population, 2024 · World Bank WDI, IP.PAT.RESD and SP.POP.TOTL · retrieved 17 Oct 2026

PART 2 OF 4 · THE REVERSAL

Korea was standing right here.

In 1980 the two countries filed almost the same number of patents.

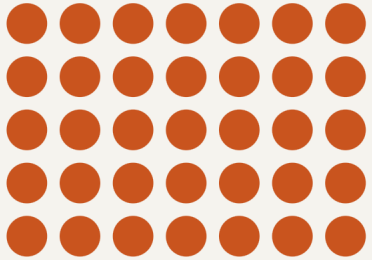
1.8×

KOREA'S ENTIRE LEAD OVER MEXICO IN 1980

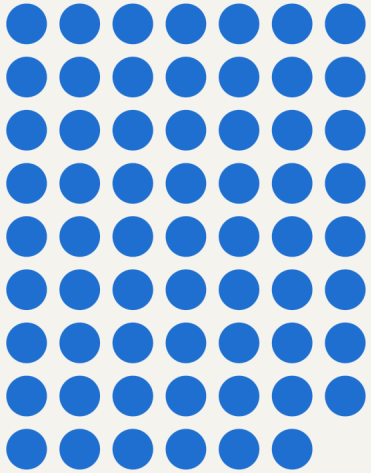
KEY

One dot is twenty patents. Same scale for both countries, and the same scale is unusable by 2024.

1980 · 1 DOT = 20 PATENTS



MEXICO 704



KOREA 1,241

so this gap is not ancient.
it was built inside one career.

Patent applications, residents, 1980 · World Bank WDI, IP.PAT.RESD · retrieved 17 Oct 2026

PART 2 OF 4 · THE REVERSAL

Then one of them started climbing.

Patents filed each year, 1980 to 2024.

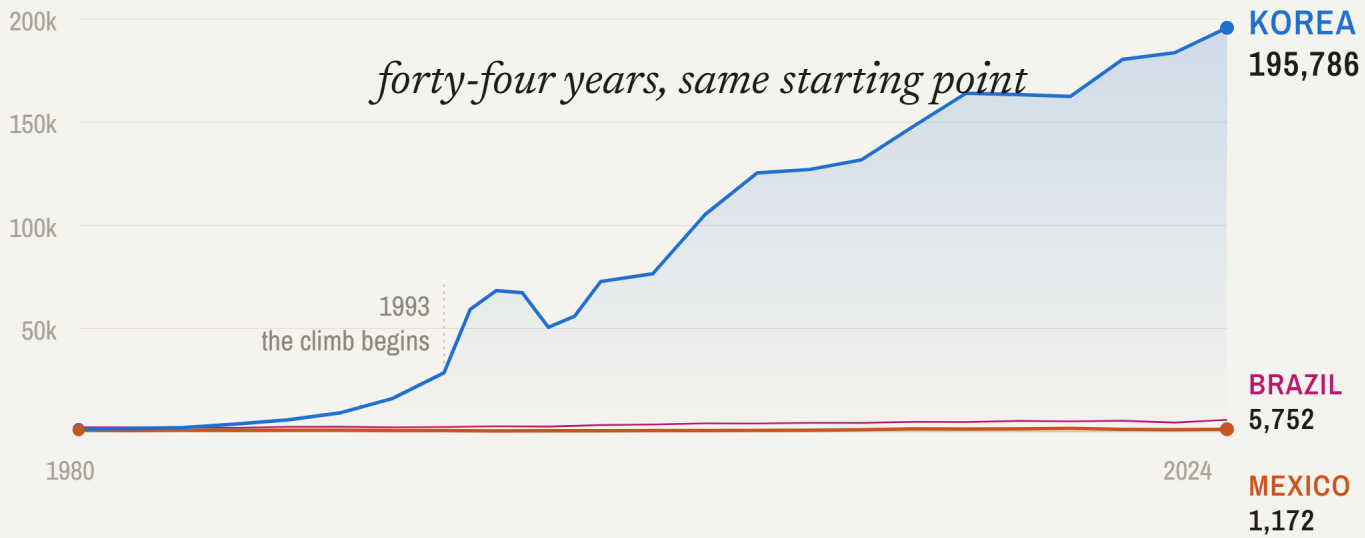
×158

KOREA'S FILINGS OVER 44 YEARS.
MEXICO'S: ×1.7

KEY

All three lines begin together in the bottom left corner. Mexico and Brazil never leave the floor, so their lines sit on top of each other.

PATENT APPLICATIONS PER YEAR



Patent applications, residents, 1980–2024 · World Bank WDI, IP.PAT.RESD · retrieved 17 Oct 2026

PART 2 OF 4 · THE REVERSAL

And it is not hidden. It is a line in the budget.

Research spending as a share of the economy.

0.3%

MEXICO'S RESEARCH SPENDING. KOREA'S: 4.9%

KEY

One dot is one tenth of one percent of the economy spent on research. Guatemala's is hollow: it rounds to zero.

R&D SPENDING, % OF GDP



Korea decided to buy this, every year, for forty years.

R&D expenditure, % of GDP · World Bank WDI, GB.XPD.RSDV.GD.ZS · latest year per country, 2022 to 2024 · retrieved 17 Oct 2026

PART 2 OF 4 · THE REVERSAL

Because almost nobody is paid to do it.

Researchers in R&D per million people.

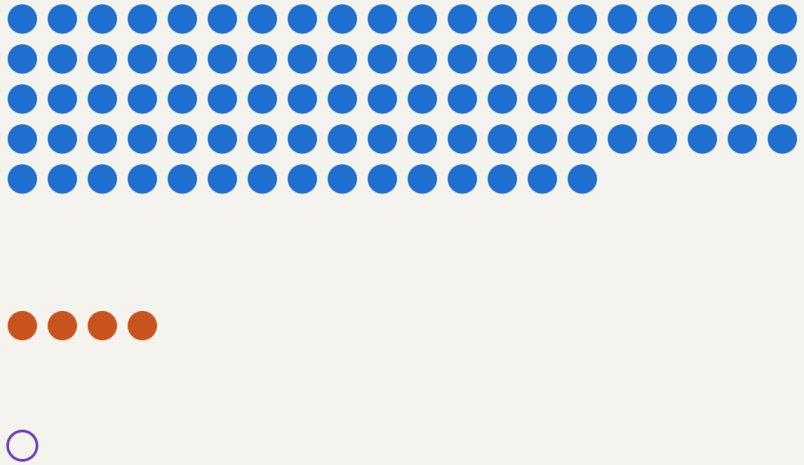
1.4

RESEARCHERS PER MILLION IN GUATEMALA. KOREA'S: 9,472

KEY

One dot is one hundred researchers per million people. Guatemala's is hollow: it is less than one.

1 DOT = 100 RESEARCHERS PER MILLION



SOUTH KOREA 9,472

MEXICO 438

GUATEMALA 1.4

1.4 per million is about 26 people in the whole country

Researchers in R&D per million people · World Bank WDI, SP.POP.SCIE.RD.P6 · Korea and Guatemala 2023, Mexico 2024 · retrieved 17 Oct 2026

This is not a mystery. It is a purchase nobody made.

Knowing how to make hard things is not a national character trait. It is bought, slowly, on purpose.

everything before this slide was evidence. everything after it is what to do.

PART 3 OF 4 · THE BILL

Sixty-four years later, here is the bill.

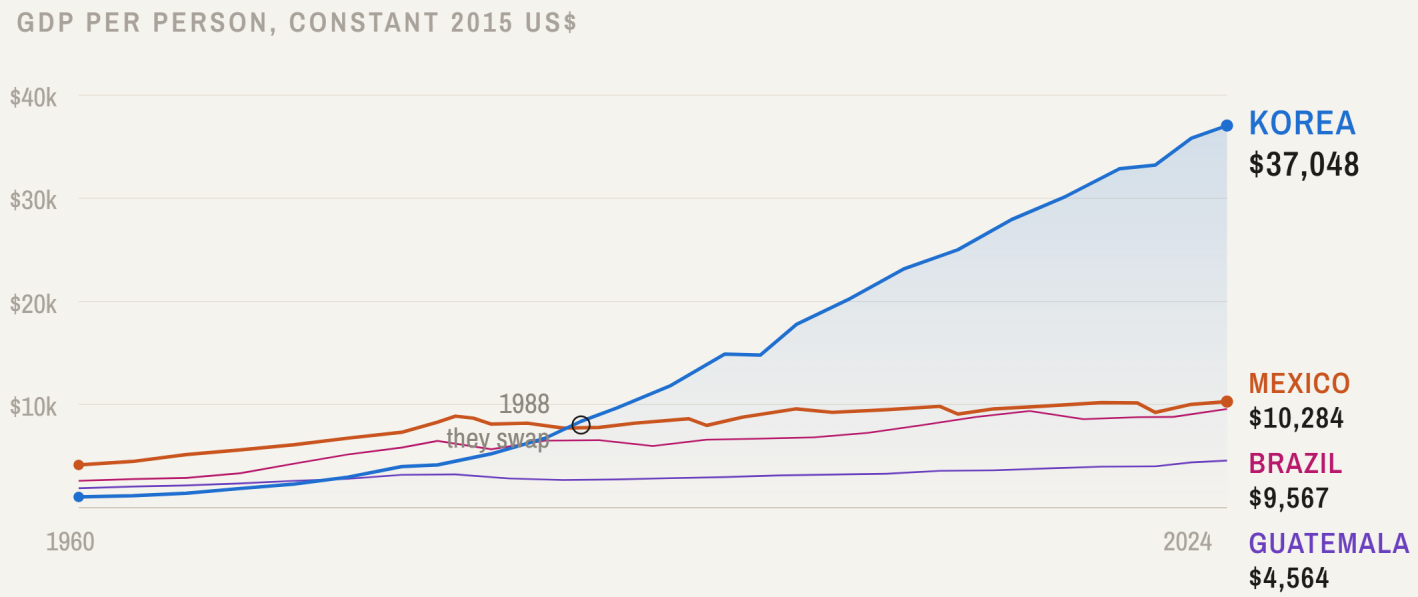
Income per person, 1960 to 2024.

3.6×

KOREA'S INCOME PER PERSON AGAINST MEXICO'S, 2024

KEY

Constant 2015 US dollars, so this is real income and not prices. The two lines cross in 1988.



in 1960 Mexico was exactly four times richer than Korea

Income per person, constant 2015 US$ · World Bank WDI, NY.GDP.PCAP.KD · retrieved 17 Oct 2026

PART 3 OF 4 · REASON ONE

The big cities took, and sent nothing back.

People, money and talent flowed in. Jobs and roads never flowed out.

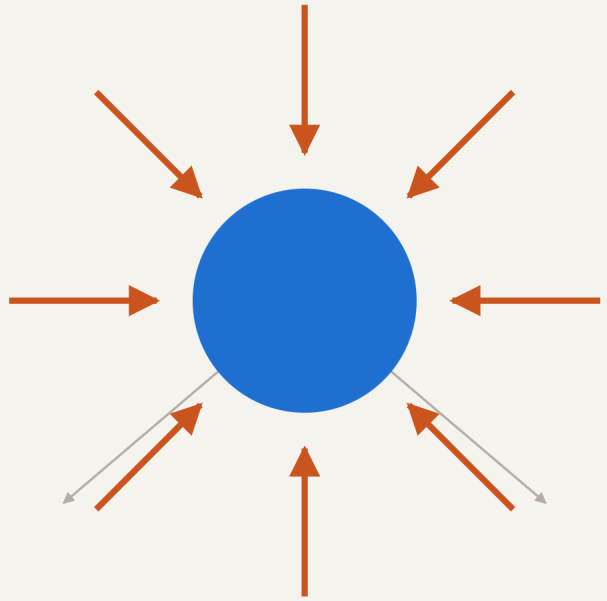
Backwash

MYRDAL, 1957 · WHAT HAPPENS WHEN GROWTH CONCENTRATES INSTEAD OF SPREADING

KEY

The blue circle is the capital. Thick arrows are what it pulls in, thin arrows what it gives back. This is a concept, not data.

THE GROWTH POLE, IN PRACTICE



PULLED IN

GIVEN BACK

economists call this backwash. everyone else calls it leaving.

Growth pole theory: Perroux 1955 · Myrdal 1957 · Hirschman 1958 · diagram is illustrative, not to scale

PART 3 OF 4 · REASON TWO

Hiring someone openly costs extra.

So firms stay small, stay hidden, and never learn to grow.

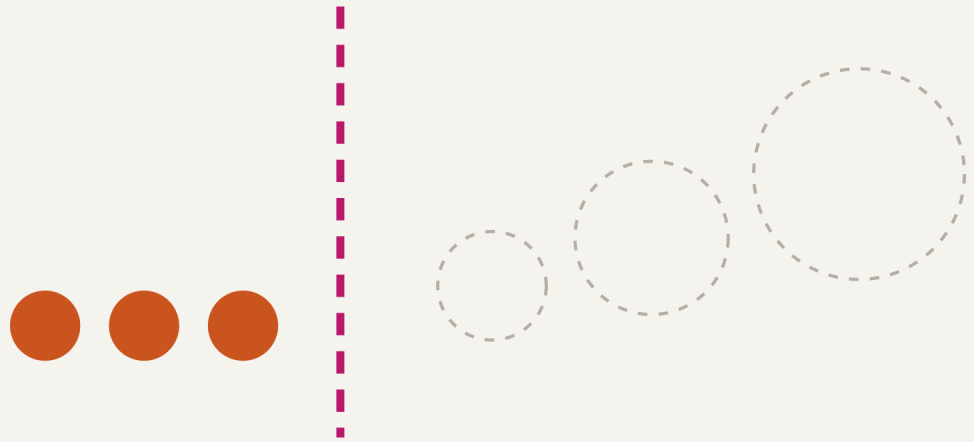
Payroll taxes buy invisibility.

MECHANISM AFTER SANTIAGO LEVY

KEY

Solid circles are the firm today. The dashed line is the moment the state can see it. Hollow circles are the growth that never comes.

WHY FIRMS NEVER GET BIG



STAYS SMALL

NEVER HAPPENS

about half the workforce lives to the left of that line – a figure we still have to pull ourselves

Mechanism after Santiago Levy · diagram illustrative · exact informality rate UNSURE until pulled from ILOSTAT, see backup slide

PART 4 OF 4 · WHAT WE DO

This has already been done. Here. On purpose.

Three cases from this region, each one engineered by a state that did one narrow thing for a long time.

None of it was luck.

ALL OF IT WAS POLICY

CASES

Fundación Chile · CINDE Costa Rica · Medellín social urbanism, from our International Cities session.

THREE PRECEDENTS, SAME REGION

CHILE

Learned to farm salmon, then handed it to firms

COSTA RICA

Went from bananas to medical devices in one generation

MEDELLÍN

Built cable cars into the hillsides the city had written off

PART 4 OF 4 · PROGRAMME ONE

Learn one new industry at a time.

Fund an agency that proves a new product works, hands it to firms, and then leaves.

Two reasons are geography. One is policy.

THE IDB CAN ONLY PAY FOR THE THIRD

KEY

Blue is public money, orange is private firms. The agency's whole job is to make itself unnecessary.

WHO CARRIES THE RISK, OVER TIME



factories are leaving China right now. they have to land somewhere.

Product life cycle, Vernon 1966 · footloose firms, from our Industrial Location session · precedent: Fundación Chile · diagram illustrative, costing UNSURE

PART 4 OF 4 · PROGRAMME TWO

Pay for the two steps that always get skipped.

Perroux's growth pole has five stages. Two of them fail by default.

Medellín's cable cars were step five.

THEY WERE NEVER A TRANSPORT PROJECT

KEY

The faded stages happen without us. The two marked in orange are the ones the money is attached to.

HOW A GROWTH POLE IS SUPPOSED TO WORK

01
A big factory arrives

02 · PAY FOR THIS
Local firms learn to supply it

03
Skills spread around

04
It keeps going on its own

05 · PAY FOR THIS
Roads and signal reach the towns outside

Perroux's five-stage growth pole process, from our Core-Periphery session · case: Medellín · costing UNSURE

PART 4 OF 4 · PROGRAMME THREE

Take the wall down.

Pay for health and pensions from general tax, not from wages.

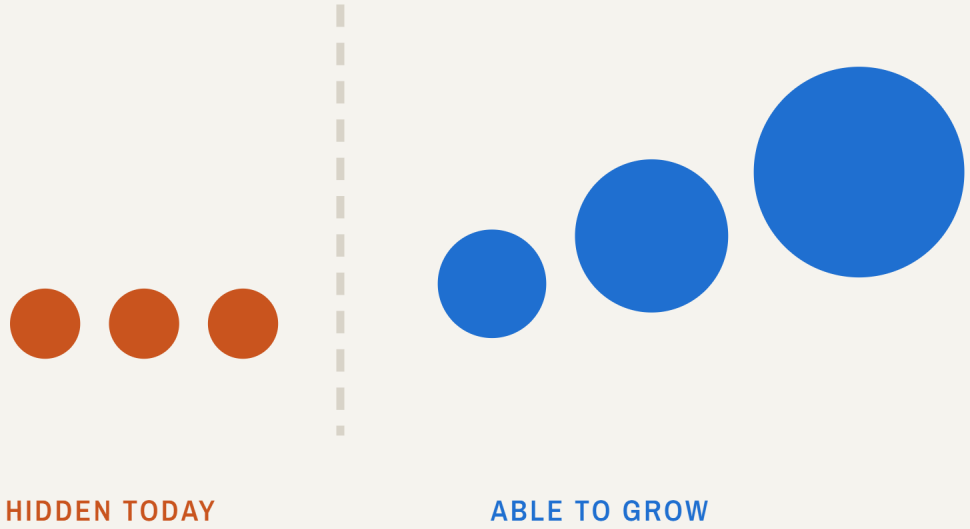
This costs money before it saves money.

SAY IT BEFORE ANYONE ASKS · IT IS ALSO THE ONLY ONE THAT PAYS BACK INSIDE ONE TERM

KEY

Same firms as four slides ago. The dashed line is the cost of being seen, faded here because we are removing it.

THE SAME PICTURE, WITH THE WALL REMOVED



the wall is a policy choice, not a fact of life

Diagram illustrative · mechanism after Santiago Levy · precedents to verify: Brazil SIMPLES, Mexico RIF · costing UNSURE

PART 4 OF 4 · SEQUENCING

Governments last four years. These do not.

So the order is chosen for political survival, not for analytical elegance.

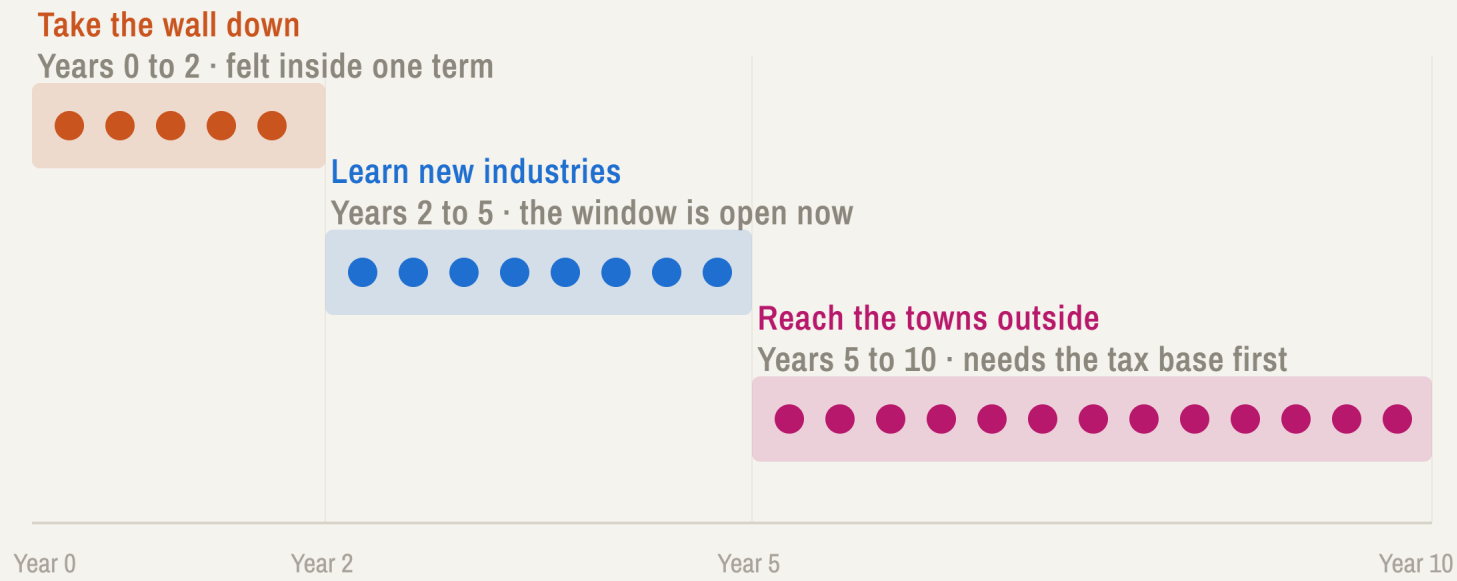
Each one buys the political room for the next.

WHICH IS WHY THE CHEAP, VISIBLE ONE GOES FIRST

KEY

Each band begins only once the one before it has shown something. The bands are sequencing, not costings.

TEN YEARS, THREE OVERLAPPING PROGRAMMES



a plan that ignores the electoral calendar never happens

Timeline illustrative · costings UNSURE, to be built from IDB project data and national budgets

PART 4 OF 4 · ACCOUNTABILITY

Three counts. Published every year.

If the IDB funds this, these are the numbers that go on the record.

We are not promising a growth number.

NO COMPARABLE SUBREGION HAS RUN THIS PACKAGE BEFORE

KEY

Solid is where we are. Hollow is where we are aiming. Shape only: the target levels are not set yet.

SOLID = TODAY · HOLLOW = WHERE WE ARE AIMING



How many patents we file

Against Chile and Brazil, not against Korea



How far the poorest region trails

Income per person, richest region against poorest



How many workers are on the books

Firms pass ten employees

a number we cannot defend is worth less than an honest blank

Baselines to be set from World Bank, ILOSTAT and national accounts · counters illustrative · target levels UNSURE

Fund the capability agency. In one country. This year.

The factories are choosing where to land right now. In five years this ask is worth less.

sixty-four years is long enough to wait for the second attempt

What is solid, what is not, and how this was made

VERIFIED, WITH CODE

All figures from World Bank WDI, pulled 17 Oct 2026: patents IP.PAT.RESD, population SP.POP.TOTL, researchers SP.POP.SCIE.RD.P6, R&D GB.XPD.RSDV.GD.ZS, income NY.GDP.PCAP.KD.

Guatemala 8 patents. Korea 195,786. Per million: 0.4 against 3,783. Every ratio recomputed from the raw series, not quoted from anywhere.

UNSURE — NOT PULLED YET

Informality rate. Chiapas income and poverty. Urban primacy share. Every costing in the three programmes.

None of these appears as a number on any slide. Sources identified: ILOSTAT, INEGI, CONEVAL, IDB project data.

WHAT THIS CANNOT TELL US

Patents measure registered invention, not all ingenuity, and firms often file abroad instead.

R&D years differ by country, 2022 to 2024. Korea's path had a geopolitical sponsor and an authoritarian state. One country is not a counterfactual for another.

USE OF GENERATIVE AI

We acknowledge the use of Claude (Anthropic, https://claude.ai) to locate sources, retrieve and recheck the World Bank series with code, and draft and design these slides. The prompts included asking for three story angles, asking it to rebuild the deck around patents rather than income, and asking it to verify every figure before use and mark anything it could not. The outputs were a basis for slides we revised; the diagnosis and the recommendation are ours.

Frameworks: Prebisch & Singer 1950 · Perroux 1955 · Myrdal 1957 · Vernon 1966 · Hausmann, Rodrik & Velasco · Levy on informality · course sessions on Core-Periphery, Industrial Location, Economic Complexity, International Cities